

Fig. 2: Use of Saber in the V&V cycle

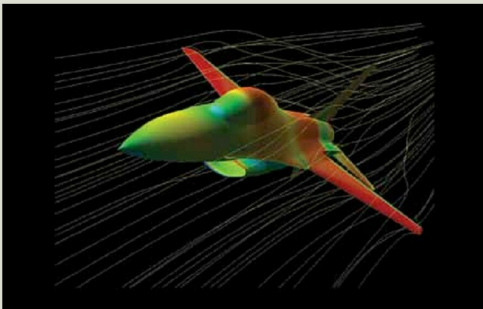


Fig. 3: Aerodynamic testing of an Airplane

installed switching and protection devices and the power users.

Modelling the systems gives an insight into the accuracy of hardware performances at both the equipment terminal and at the temporal and frequency domain. Simulation also serves to establish the integration at network level between the model library, model interface and solver parameters, thus making the whole process seamless and easier to troubleshoot and maintain.

Several levels of modelling are required to reach the objectives of simulation. In the order of increasing complexity, they would be architectural, functional, behavioural and components. The behavioural model of simulation is used for power-quality study of the network including transient-level voltage, harmonic distur-

tion and voltage ripple. The functional model of simulation mainly deals with the network stability studies along with validation of network configuration logics.

Simulation activity

Functional. The major sources of failure generation at the network level can be narrowed down to the following:

1. Open cable
2. Short circuit
3. Erroneous information signal
4. Contact stuck open or close
5. Power distribution centres
6. Power converters
7. Electric loads

Functional-level modelling aims to integrate voltage generators, power distribution centres, power converters and electric loads and others at the

network level to increase the overall system dependability.

Simulation in this modelling follows the following algorithm. The process starts with the construction of a test bench, which is basically a virtual environment used to verify the correctness or soundness of a design or model. The issues that pop up due to the convergence of multi-physics are taken into account and solved as per requirement before the model is simulated. After simulation, post-processing is done. Simulation results include the results of simulating the entire electrical network at a functional level along with finding out the impact of possible failures on the electrical network.

Saber, Synopsys' simulation tool, offers users the ability to use both analogue and digital solvers to simulate and work on other features while simulating. It also allows users to use other tool models through the ability to import C-code capabilities.

Power quality. Power-quality studies at network level deal with the transient voltage levels on distribution bus bars including extreme conditions like heavy load switching, high power demand from dynamic loads and bus-bar transfers. They also study steady-state voltage harmonic content on distribution bus bars with special focus on total harmonic distortion and harmonic spectrum. Major issues that crop up are the integration of non-linear models and the ability to start the simulation from any steady-state condition.

Failure-mode effects analysis (FMEA). Modern aerospace systems are dependent on electrical control. Design teams must understand how electrical failures affect system performance. A good simulator can be used to simulate and analyse systems under electrical fault conditions. A matrix of faults representing electrical device failures (e.g., system shorts or opens) is automatically created to evaluate system performance during each fault condition. Fault reports are automatically generated and used to